If $\gcd(j,n) = \frac{n}{d_i}$, then $k = \frac{j}{n/d_i}$ is relatively prime to d_i and is in the range $1 \le k \le d_i$, i.e. k is a totative of d_i. The number of such elements k is $\phi(d_i)$, by definition, so C_{d_i} also has $\phi(d_i)$ elements: take all these values of k and multiply them by $\frac{n}{d_i}$. Since every element j with $1 \le j \le n$ is in exactly one C_{d_i}, it follows that the sum of the number of elements in the C_{d_i}'s is equal to n, which is simply (7.1). ∎

Example. Let's do this partitioning when $n = 14$. The divisors of 14 are 1, 2, 7, and 14. The set C_1 consists of those numbers j from 1 to 14 with $\gcd(j,14) = \frac{14}{1} = 14$, so just 14. Similarly, C_2 consists of those numbers j with $\gcd(j,14) = \frac{14}{2} = 7$, which is just 7. Continuing on this way, we get

$$C_1 = \{14\}, \qquad C_2 = \{7\}, \qquad C_7 = \{2,4,6,8,10,12\},$$
$$C_{14} = \{1,3,5,9,11,13\}.$$

Note that for each divisor d of 14, the number of elements in C_d is $\phi(d)$, as claimed in the proof.

7.3 Problems

(1) Find, with proof, all integers x such that $6x \equiv 4 \pmod{28}$.

(2) Give a good algorithm for finding multiplicative inverses modulo m, provided they exist. That is, given a and m with $\gcd(a,m) = 1$, explain how to compute $\frac{1}{a} \pmod{m}$. Use your algorithm to compute $\frac{1}{14} \pmod{107}$. (Hint: Think about the Euclidean algorithm and Bézout's Lemma.)

(3) Prove that if p is prime, then the only elements of $(\mathbb{Z}/p\mathbb{Z})^\times$ that are congruent to their multiplicative inverses modulo p are $\pm 1 \pmod{p}$, i.e. these are the only $x \in (\mathbb{Z}/p\mathbb{Z})^\times$ such that $x \equiv \frac{1}{x} \pmod{p}$. Find an m such that there are more than two elements of $(\mathbb{Z}/m\mathbb{Z})^\times$ that are congruent to their own inverses. Show that for any positive integer n, there is an integer m (depending on n) such that at least n elements of $(\mathbb{Z}/m\mathbb{Z})^\times$ are congruent to their own inverses.

(4) Prove that if $\gcd(a,b) = g$, then

$$\phi(ab) = \frac{g\phi(a)\phi(b)}{\phi(g)}.$$

(5) Prove that for any integer $n > 1$, we have

$$\sum_{\substack{1 \le i \le n \\ \gcd(i,n)=1}} i = \frac{n\phi(n)}{2}.$$

(6) Prove that if $d \mid n$, then $\phi(d) \mid \phi(n)$.

(7) Find all numbers n such that $\phi(n) = 36$. Prove that you found all of them.

(8) Prove that for every positive integer k, there are only finitely many positive integers n such that $\phi(n) = k$.

(9) Suppose f is a multiplicative function, and let

$$g(n) = \sum_{d|n} f(d).$$

Prove that g is a multiplicative function.

(10) Let ε be any positive real number. Prove that there is some positive integer n, depending on ε, such that $\frac{\phi(n)}{n} < \varepsilon$. (This is probably too hard in general, given what we have discussed so far. But you should be able to do it when $\varepsilon = \frac{1}{5}$, so do that if you can't do the general case. The construction for the general case isn't significantly different from what you'll probably do for $\frac{1}{5}$, but the problem is in proving that it gets arbitrarily small.)

*$15 \equiv 1 \bmod 14$, so it just becomes $1 \cdot x$

T+P Week 7: Mod mult. and div. Alexander

7.1: By prop 7.3, $2 \cdot 3x \equiv 2 \cdot 2 \bmod 2 \cdot 14$, so (gcd of 2 and 28)

$3x \equiv 2 \bmod 14$. 3 is a unit mod 14,

so $x \equiv \frac{2}{3} \bmod 14$. (this is a step backwards...)

$3x \equiv 2 \bmod 14$, so $15x \equiv 10 \bmod 14$,

or just $\boxed{x^* \equiv 10 \bmod 14.}$ 10, 24, 38, 52...

7.2: We want to find n such that $\frac{1}{a} \equiv n \pmod m$,

or $1 \equiv an \pmod m$, or $1 - an = km$ for

some k. $1 = an + km$. Since $\gcd(a,m) = 1$,

this is an example of Bezout's Lemma

where $na + km = 1$. So n and k exist.

Now let's find ~~both of~~ them. Start with

$1 \cdot a + 0 \cdot m = a$ and $0 \cdot a + 1 \cdot m = m$. Then

subtract one equation by the other

based on whether $m > a$ or $a > m$, such

that the result is positive on right.

With the resulting equation, again

perform this action - just Euclid's Algorithm

keep going until the right side is 1,

and we have our n and k. n is our inverse!

NEXT: $14 + 0 = 14$ and $0 + 107 = 107$. $\sim 7 \cdot 14 + 1 \cdot 107 = 9$. $8 \cdot 14 - 1 \cdot 107 = 5$.

50 23. $\leftarrow$ (23) $14 - 3 \cdot 107 = 1$, $-15 \cdot 14 + 2 \cdot 107 = 4$,

7.3: $x - \frac{1}{x} = kp$ for some k. $x^2 - 1 = (x \cdot k)p = (x+1)(x-1)$.

For an element of $(\mathbb{Z}/p\mathbb{Z})^{(x)}$, or just $\pm(1 \le n \le p-1)$, to be $\equiv \frac{1}{x} \bmod p$, $(x+1)$ or $(x-1)$ must be divided evenly by p. Thus, x can only be $\equiv$ to $\pm 1 \bmod p$.

Any composite number satisfies. Take 10: 2 and 5, as well as ± 1, make 4 total.

Choose a value k such that $2^k \ge n$.

Define $m = p_1 \cdot p_2 \cdot \ldots \cdot p_k$ where all p_i's are distinct. All instances of $x \equiv \frac{1}{x} \bmod m$, or $x^2 \equiv 1 \bmod m$, are $x^2 \equiv 1 \bmod p_1 \ldots \bmod p_2 \ldots \bmod p_k$. Each prime yeilds 2 solutions for x (± 1), so by the CRT, the number of solutions with all primes multiplied is $2 \cdot 2 \cdot 2 \ldots \cdot 2$ (k times), or 2^k. This suffices because $n \le 2^k$.

7.4: By Th. 7.11, $\phi(n) = n \prod_{p \mid n,\ p \text{ is prime}} (1 - \frac{1}{p})$, and this applies to $n = a, b$, and g. So $\phi(a)\phi(b) = ab \cdot \prod_{p \mid a} (1 - \frac{1}{p}) \prod_{p \mid b} (1 - \frac{1}{p})$ (cont.)

(Prime Factorizations of a, b)

*$P(n)$ = prime factorization of n. $\$$ This would be the LCM... if powers mattered. $\#$ It actually doesn't matter what power p is raised to.

7.4 (cont.): The prime factorizations* of a and b together are equivalent to $P(a) \cup P(b) \cdot P(a) \cap P(b)$, i.e. each prime power factor$^{\#}$, with the extra primes shared from the $\gcd(a,b)$. So the product $\phi(a)\phi(b)$ is also $ab \cdot \prod_{p \in P(a) \cup P(b)\ \$} (1-\frac{1}{p}) \cdot \prod_{p \in P(a) \cap P(b)} (1-\frac{1}{p})$.

Note that $ab \cdot \prod_{p \in P(ab)} (1-\frac{1}{p})$ is just $\phi(ab)$, and that $\prod_{p \in P(g)} (1-\frac{1}{p})$ is just $\phi(g)$ divided by g. Substituting: $\phi(a)\phi(b) = \phi(ab) \cdot \frac{\phi(g)}{g}$, or $\phi(ab) = \phi(a)\phi(b) \cdot \frac{g}{\phi(g)} = \boxed{\frac{g\phi(a)\phi(b)}{\phi(g)}}$. ■

7.5: For each unit i, $n-i$ is also a unit in $(\mathbb{Z}/n\mathbb{Z})^{\times}$ by the Euclidean Algorithm. Each of these pairs sum to n, and there are $\phi(n)/2$ total pairs. Multiply these to get $\boxed{\frac{n\phi(n)}{2}}$ as the sum of totatives.

7.6: Say that $n = d \cdot k$ for some k. If $\phi(d) \mid \phi(n)$, then $d \prod_{p \mid d} (1-\frac{1}{p}) \mid dk \prod_{p \mid dk} (1-\frac{1}{p})$. ~~$d \cdot$ something must divide $d \cdot k \cdot$ something else, because both are multiples of d!~~ All primes dividing d also divide $k \cdot d$, so the first product divides the second. d also divides dk, so $\phi(d) \mid \phi(n = dk)$

7.7: $\phi(n) = \prod(p_i^{e_i} - p_i^{e_i-1}) = 36$ for each prime factor of n, p_i. 36 factors: 1, 2, 3, 4, 6, 9, 12, 18, 36

$1 = 2^1 - 2^0$, $2 = 2^2 - 2^1$, $3^1 - 3^0$; 3 = Nothing,

$4 = 2^3 - 2^2$, $5^1 - 5^0$, $6 = 3^2 - 3^1$, $7^1 - 7^0$; 9 = Nothing,

$12 = 13^1 - 13^0$, $18 = 3^3 - 3^2$, $19^1 - 19^0$, $36 = 37^1 - 37^0$.

This yields 108, 76, 57, 63, 37, 74, 114, 126.

These values come from all possible products of the primes found above where their consecutive power differences also multiply to 36. This was annoying...

7.8: $\phi(n) = \prod(p_i^{e_i} - p_i^{e_i-1}) = \prod p_i^{e_i-1}(p_i - 1)$. For each $p_i \mid n$, $(p_i - 1)$ must divide $\phi(n)$. So $(p_i - 1) \leq \phi(n)$. Since $\phi(n) = k$, $p_i - 1 \leq k$, or $p_i \leq k+1$.

This is a finite bound for all possible primes that make up $k = \phi(n)$, so $\phi(n)$ is finite too.

7.9: Trying to prove $g(mn) = g(m)g(n)$, or (where $d' = f(d)$)

$(d_1' + d_2' \ldots + d_i' : d \mid mn) = (d_1' + \ldots d_i' : d \mid m) \cdot (d_1' + \ldots d_i' : d \mid n)$.

f is multiplicative, and so is $\sum$, so $g(mn) = \sum_{d \mid mn} f(d) = \sum_{d_i \mid m} \sum_{d_j \mid n} f(d_i d_j)$

$= \sum_{d_i \mid m} \sum_{d_j \mid n} f(d_i) f(d_j)$. This factors into $\left(\sum_{d_i \mid m} f(d_i)\right)\left(\sum_{d_j \mid n} f(d_j)\right)$,

which is just $g(m) \cdot g(n)$. so g is multiplicative.

7.10: Start with $\varepsilon = 1/5$. Prove there exists n where $\frac{\phi(n)}{n} < \frac{1}{5}$, or where $\frac{n \cdot \prod_{p|n}(1-\frac{1}{p})}{n} = \prod_{p|n}(1-\frac{1}{p}) < \frac{1}{5}$. The first instance where this product equals $n < \frac{1}{5}$ is $1/2 \cdot 2/3 \cdot 4/5 \cdot 6/7 \cdot 10/11 \cdot 12/13 = 0.191808192\ldots < 0.2$

The least number with all these prime factors is $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 = \boxed{210210}$. So an n exists for $\varepsilon = 1/5$. General case? We already have $\prod_{p|n}(1-\frac{1}{p}) < \varepsilon$, so all we have to show is that the product approaches 0 for more primes that divide even larger n's.

Reciprocal: $\frac{n}{\phi(n)} = \prod_{p|n}(1-\frac{1}{p_i})^{-1}$. $\frac{1}{1-\frac{1}{n}} = 1 + \frac{1}{n} + \frac{1}{n^2} + \ldots$ by infinite sum formula, so the big product $= (1 + \frac{1}{p_1} + \frac{1}{p_1^2} \ldots) \cdot (1 + \frac{1}{p_2} + \frac{1}{p_2^2} \ldots) \cdot \ldots \cdot (1 + \frac{1}{p_i} + \frac{1}{p_i^2} + \frac{1}{p_i^3} \ldots)$.

By the Fundamental Th. of Arithmetic, this evaluates to the sum of all reciprocals of numbers less than p_i^k (all possible prime factorizations). As k (the size of each sum) approaches ∞, so does the possible prime factorizations. Their reciprocals, then, will approach 0. So our original $\frac{\phi(n)}{n}$ approaches 0 as well.